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Towards explainable artificial intelligence in deep vision-based odometry

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ABSTRACT

Visual Odometry (VO) is a crucial process for estimating camera motion in real-time based on visual information captured. The emergence of deep learning has significantly transformed VO and Explainable Artificial Intelligence (XAI) in Deep Vision-Based Odometry. This survey paper explores the latest advancements in VO facilitated by deep learning methods, focusing on explainability and interpretability. It provides an overview of state-of-the-art deep learning techniques and dissects each model into its elemental building blocks to understand their explainable and interpretable aspects. The survey also highlights research gaps in optical flow robustness, occlusion and dynamic objects, real-time processing, drift correction, semantic awareness, and sensor integration. The aim is to catalyze future innovations in deep learning-based VO and stimulate dialogue about potential directions for the next wave of research, emphasizing explainability and interpretability as integral components of advanced systems.

1. Introduction

Self-localization is the process of estimating the position, orientation, and motion of a platform (e.g., a robot or a vehicle) in its environment without relying on external sources of information, such as GPS or other positioning systems. Self-localization is a critical capability for autonomous agents, as it allows them to navigate and operate in unstructured or dynamic environments where external sources of information may not be available or reliable. Self-localization can be performed using a variety of sensors and algorithms. Typical self-localized sensors include cameras, LIDAR, sonar, radar, inertial measurement units (IMUs), and wheel encoders. Algorithms for self-localization range from simple dead reckoning (using motion measurements to estimate position) to complex probabilistic methods such as the Extended Kalman Filter (EKF) or the Particle Filter. Self-localization accuracy depends on several factors, including the sensors' type and quality, the localization algorithm's complexity and sophistication, and the environment in which the agent operates. Self-localization can be particularly difficult in challenging environments such as indoor or urban settings due to sensor noise, occlusions, and signal interference. This comprehensive survey paper has dissected most visual odometry methods into their fundamental building blocks. We aim to illuminate the diverse techniques employed in different manners and combinations, primarily emphasizing explainability and interpretability as integral facets of advanced systems. This approach is designed to empower readers to gain a detailed understanding of these models and to enable them to identify suitable avenues for future enhancements within each building block, ultimately refining the accuracy of pose estimation. Additionally, we have outlined several potential research opportunities, including addressing robustness issues related to optical flow models in varying lighting conditions, tackling long-term

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consistency concerns and drift correction, handling challenges related to occlusions and dynamic objects, optimizing real-time performance, enhancing computational efficiency, integrating semantic awareness into visual odometry, and exploring integration possibilities with other sensors. Consequently, this survey is dedicated to identifying research gaps by scrutinizing model architectures and highlighting the building blocks that, to our knowledge, have not yet been thoroughly explored.

The main contributions to the readers provided by this paper can be summarized as:

- A comprehensive overview of recent advances in visual odometry, focusing on deep learning models.
- An in-depth model comparison, highlighting explainability and interpretability aspects.
- A summarization of critical research gaps, especially in the context of Explainable AI (XAI) in VO.
- A discussion on advances in XAI techniques for visual odometry, allowing transparent models.
- A summary of key challenges in visual odometry, shedding light on areas that require further attention and innovation to enhance the field.
- An overview of practical implications of visual odometry advancements, demonstrating their relevance and potential impact in real applications, such as self-driving cars and robots.

The rest of this paper is organized as follows. Section II provides a background on self-localization; Section III introduces visual odometry and a classification of its methods. Section IV discusses deep learning usage in visual odometry with an understanding of recent advancements. Section V gives a decomposition of the building blocks of pose estimation methods. Section VI discusses the famous datasets widely used in model training and comparison analysis. Section VII provides a comparison between models and discusses the challenges of each. Section VIII presents a short overview of visual odometry usage in the industry, and finally, section IX recaps the research gaps and concludes the survey.

2. A background on self localization

This section briefly reviews different aspects of self-localization, including its importance, applications in different industries, various environments, and associated limitations. Self-localization, or the ability to determine one's position and orientation in each environment, is essential for many tasks in robotics, autonomous vehicles, and other fields that involve autonomous navigation or interaction with the physical world. Here are some reasons why self-localization is important. Self-localization is essential for robots and autonomous vehicles to navigate complex environments, plan trajectories, avoid obstacles, and reach their destination more efficiently. It is also important for building accurate maps of the environment, object recognition, task planning, and safety. Self-localization allows robots to understand better the relationships between different objects and landmarks in the environment, which helps to create more detailed and accurate maps. Object recognition is particularly important in applications such as robotic grasping, where the robot must know the precise location and orientation of the object it tries to manipulate. Task planning helps robots to plan tasks more effectively, while safety is essential for ensuring the safety of autonomous systems. Developing and investigating methods for Self-Localization / Odometry systems have made great strides in robotics [1]. A crucial problem for autonomous navigation, path planning, object identification, and obstacle avoidance platforms is the capacity to continually estimate the robot's ego-motion, including position and orientation over time. For an autonomous system, Unmanned Ground Vehicles (UGV) or Unmanned Aerial Vehicles (UAV), determining location and orientation throughout the traversing or flight is a critical problem. This ability becomes more significant when the operation is outdoors, and no such environmental configurations as indoors can be done. The wide range of natural and unnatural diversity in the land, including land features such as mountains, plains, seas, and rivers, or great diversity in various buildings and streets, makes it more challenging to accurately determine the robot's location.

2.1. Indoor vs. outdoor localization

Indoor and outdoor localization are two different aspects of self-localization, and they have their unique challenges and applications. More specifically, Indoor localization refers to determining an agent's location inside a building or a closed environment. It is a critical requirement for various indoor applications, such as indoor navigation, asset tracking, and robotics. Indoor localization is a challenging task due to the limitations of the GPS, which cannot penetrate buildings' walls and roofs. Therefore, different localization techniques such as Wi-Fi fingerprinting, Bluetooth low energy, ultrasonic, and infrared are used to determine an agent's indoor location. On the other hand, outdoor localization refers to determining an agent's location in an open environment, such as a park, a city, or a rural area. Outdoor localization is relatively easier than indoor localization since the agent can accurately determine its location using global positioning systems (GPS). However, outdoor localization still presents some challenges. This paper focuses on outdoor localization, including methods, applications, and data sources.

2.2. Global Navigation Satellite System-based methods

Global Navigation Satellite System (GNSS) based methods, such as GPS, GLONASS, and Galileo, are widely used for outdoor localization due to their accuracy and reliability. However, they have limitations, such as signal blockage, multipath error, signal jamming, cost and power consumption, and limited indoor use. Signal blockage occurs when the GNSS signals are blocked or attenuated by buildings, trees, or other obstacles. A multipath error occurs when the GNSS signals reflect off buildings or other surfaces, creating multiple signal paths. Signal jamming occurs when the GNSS signals are vulnerable to intentional or unintentional

interference, such as jamming or spoofing. Cost and power consumption are also limitations for low-cost or power-constrained applications. Despite these limitations, GNSS-based methods remain the most widely used and accurate outdoor localization techniques. To mitigate these limitations, other complementary methods, such as inertial navigation systems, cellular network-based localization, or visual odometry, can be used in conjunction with GNSS to improve localization accuracy and reliability. GNSS, like GPS, is a well-known localization mechanism in various autonomous system areas. Typically, the GPS signal and an IMU are used to manage this. Any user with a GPS receiver may obtain location data via GPS with meter-level precision. However, this method is not only ineffective in an indoor environment but also vulnerable to failure in an outdoor environment because of GPS jamming and multipath attenuation or even the absence of GPS signals during planetary robotic explorations and landing on designated landing sites or navigating autonomously to a destination.

2.3. Self-localization (GNSS-denied) techniques

Many different localization methods are proposed to address the current issue with GNSS-based methods. Each technique is designed for a specific environmental condition and input data type. This would enable autonomous systems to function in GNSS-denied areas and provide users with essential data, such as real-time 3D reconstruction maps. Fig. 1 presents a classification of self-localization techniques in the absence of a GNSS signal. The methods can be divided into single-modality and multi-modality [1].

Single-Modality: Using a single sensor as the only source of environmental information is the most used method due to the lower hardware cost. This category can be divided into 3 subcategories: Vision-based, Observation-based, and Inertial. For each subgroup, different types of sensors are used. The primary sensors are camera (RGB, event-based), LiDAR, wheel, Sonar, Radar, and Inertial Measurement Unit. Among all mentioned data sources, the RGB camera attracted the most attention due to its low cost compared to other sensors.

Multi-Modality: Some methods use multiple sources of information and proposed fusion-based methods. The favorite approach is vision and inertial fusion, mainly called Vision-Inertial Odometry (VIO). Fewer methods are proposed since these methods need more hardware installed on the system, more computational resources for processing, and fewer available public datasets. However, considering the cost of their implementation, the greater accuracy of these combined methods deserves more investigation and research. Visual odometry is a type of single-modality navigation that estimates the robot's motion (rotation and translation) and utilizes that information to locate the robot within its surroundings. Using visual markers, the onboard vision system calculates rotation and translation motion parameters between two places in time. In recent years, the development of more precise and cheap cameras and the accessibility of this sensor compared to other types of sensors have encouraged researchers to propose models designed to get the image as their only input. Therefore, the VO became more popular than other methods. Moreover, the advancement of computer vision techniques and the performance of proposed models bring more opportunities for localization applications to use cameras to estimate pose, especially in denied GPS zones or anywhere the GPS signal is unreliable. Therefore, in this paper, we only focus on VO methods.

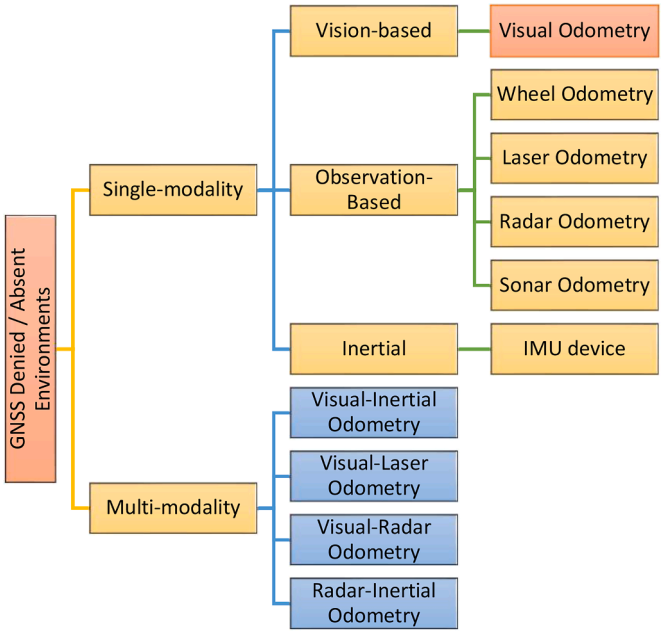


Fig. 1. GNSS denied / absent self-localization techniques.

3. Visual Odometry (VO)

The idea of VO is comparable to wheel odometry, which integrates the number of wheels turns over time to estimate the pose and motion of the vehicle incrementally. Additionally, VO calculates the position progressively by analyzing the differences in motion caused by a collection of pictures taken by the on-board camera(s). The first step in the VO pipeline is collecting a sequence of images from the camera(s). Then, extract features from the image using conventional image processing techniques or deep learning. Track the extracted features and estimate motion. The final stage is optimizing the posture estimate for a more resilient and accurate result. Fig. 2 presents the different categories of visual odometry methods. Early methods mostly used geometry-based algorithms like extracting features or appearance in consecutive images.

3.1. Geometric-based VO

Geometric-based VO hinges on analyzing the geometry of a scene captured through a sequence of images, typically obtained from a camera. By detecting and tracking specific visual features across these images, geometric-based VO algorithms estimate the camera's motion relative to its surroundings. This process involves intricate calculations of feature correspondences and subsequent triangulation to determine the camera's position and orientation in space. Geometric VO, distinguished by its reliance on mathematical models and algorithms rather than machine learning, is particularly valued for its precision and effectiveness in various applications, from autonomous vehicles to augmented reality, where accurate and reliable motion tracking is paramount. Its development and refinement continue to be a vibrant area of research, aiming to enhance robustness, accuracy, and efficiency in diverse and challenging environments. It has two key subclasses:

3.1.1. Feature-based

These methods extract distinctive features from two or more images and then match them to compute the relative motion. For feature extraction and matching, methods such as Scale-Invariant Feature Transform (SIFT), Speeded Robust Features (SURF), Oriented FAST and Rotated BRIEF (ORB), and others can be utilized. As some examples of well-known methods, SIFT detects and uses local features in images and is invariant to image scale and rotation. SURF is a detector and descriptor that is quicker to compute than SIFT and is used in real-time applications. ORB is another real-time suitable method that combines the FAST keypoint detector and BRIEF descriptor with some modifications. Visual Odometry algorithms such as PTAM (Parallel Tracking and Mapping) and ORB-SLAM utilize feature extraction methods for visual odometry. This category can further be divided into two primary categories based on how they treat the features [2]:

3.1.1.1. Sparse methods. These methods extract and use only a few distinct features from the image, such as corners or other highly distinctive areas. This approach is computationally efficient and works well in environments with lots of texture or distinct structures. The algorithm PTAM (Parallel Tracking and Mapping) is a prime example of a sparse method. Sparse techniques are also commonly connected with approaches that use features such as SIFT, ORB, and SURF.

3.1.1.2. Dense methods. These methods attempt to extract features from every part of the image, resulting in a dense map of features. These methods are more computationally expensive but can provide a more detailed understanding of the environment. They are helpful in texture-less or structure-less environments where sparse features are hard to extract. An example of this method in action is in Dense Visual SLAM systems for small-scale environments. Further to these, there are also semi-dense methods that lie in between the sparse and dense methods. They extract features from edges in the image, which typically covers a significant portion of the image but is not as computationally expensive as dense methods. LSD-SLAM (Large-Scale Direct Monocular SLAM) is one method that uses a

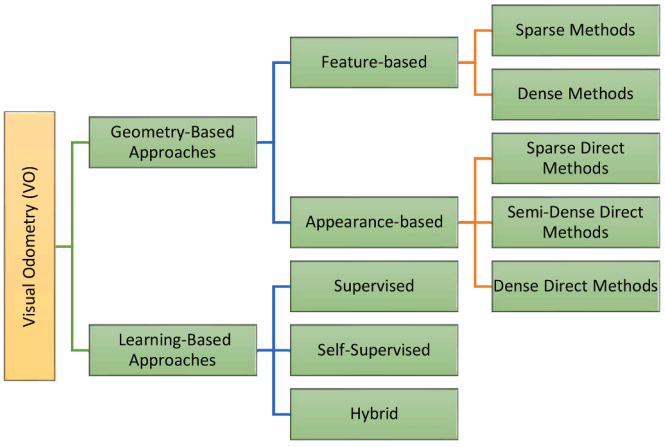


Fig. 2. Classification of the Visual Odometry (VO) techniques.

semi-dense approach, although it can also be classified under the direct or appearance-based methods.

3.1.2. Appearance-based (or direct methods)

These methods use image intensity values or color information to estimate motion. They typically do not require explicit feature extraction or matching and instead try to minimize the photometric error directly. For example, DTAM (Dense Tracking and Mapping) uses dense depth maps and works directly on pixel intensities. LSD-SLAM optimizes photometric errors and works on a semi-dense depth map. Similar to feature-based methods, appearance-based (also known as direct) methods can also be divided into categories based on the density of the used image information:

3.1.2.1. Sparse direct methods. These methods are selective about the pixels they use for photometric error minimization, typically choosing those that are easy to track or provide distinctive information. Sparse direct methods often fall somewhere between feature-based and dense direct methods in terms of computational complexity and robustness [2]. An example of a sparse direct method is Semi-Dense Visual Odometry (SDVO).

3.1.2.2. Dense direct methods. These methods use all or most of the pixels in the image for estimating the camera motion. This approach can provide highly accurate results and detailed reconstructions but at the cost of increased computational complexity. Dense direct methods can handle texture-less or low-textured environments better than feature-based methods, which do not rely on distinct features. Examples include DTAM (Dense Tracking and Mapping) and LSD-SLAM. However, LSD-SLAM can also be considered semi-dense as it uses pixel intensities along edges in the image.

3.1.2.3. Semi-dense direct methods. As mentioned, these methods use a subset of image pixels for motion estimation, typically along the edges or in regions with high gradient magnitudes. The semi-dense technique can offer a fair balance of accuracy and processing expense. LSD-SLAM, for instance, can be considered a semi-dense method [3].

3.2. Learning-based VO

3.2.1. Traditional methods

Traditional machine learning-based visual odometry represents an earlier phase in learning-assisted motion estimation techniques. In these methods, machine learning algorithms are primarily used to enhance specific aspects of the VO pipeline, such as feature detection, feature matching, or motion estimation. Unlike deep learning approaches, traditional machine learning in VO often relies on handcrafted features and classical algorithms like Support Vector Machines (SVMs), Decision Trees, or simple neural networks. These methods require explicit programming of the features to be detected in the images, and the learning algorithms then optimize these features for better motion tracking and estimation [3]. The focus is typically on improving the VO system's precision and resilience under various conditions, such as changes in lighting or motion blur. However, traditional machine learning methods in VO are generally more constrained and less flexible than the more recent deep learning approaches. We are covering the traditional ML methods for brevity in the survey.

3.2.2. Deep learning methods

In contrast to traditional ML methods, the second category of learning-based VO methods, which we will cover later, delves into deep learning. Deep learning-based VO represents a significant shift from traditional techniques, employing complex neural network architectures to learn directly from raw data. This approach often improves performance and generalization capabilities, especially in dynamic and unstructured environments. We will explore these deep learning-based methods in more detail, highlighting their unique advantages and the challenges they address in visual odometry. The geometric methods relied solely on the features humans detected. Then, by extracting them and matching them through multiple consecutive images, calculate the camera motion. However, in the learning-based methods (e.g., deep learning), all these 3 steps: feature extraction, matching, and pose calculation are carried on by learning models. Many more elements not identifiable by humans will be employed to estimate the camera's attitude in this scenario.

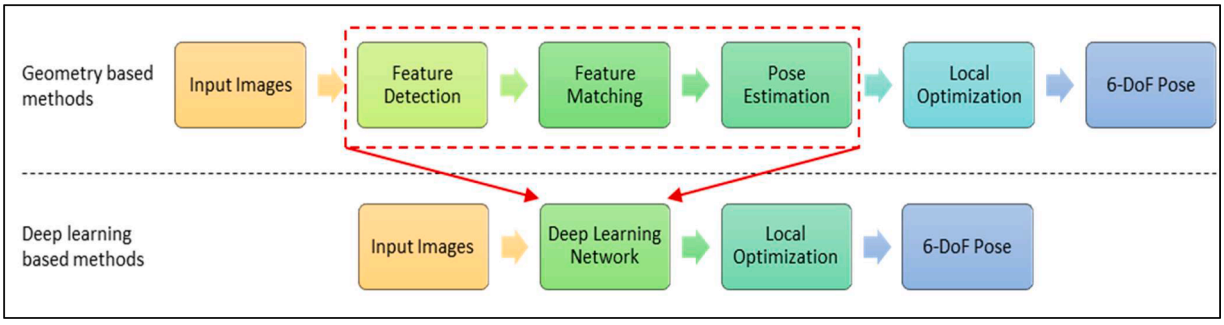


Fig. 3. . Deep learning-based vs geometry-based methods [4].

Since the previous decade, existing research has concentrated on deep learning-based methodologies. However, in recent years, almost all the research used a learning approach as the primary method, especially deep learning, which opened a new era in visual odometry. This category of methods employs machine learning techniques, especially deep learning, to estimate the motion from consecutive images. Defining clear-cut subclasses for learning-based methods is more challenging because they can vary widely in their network architectures, input data, loss functions, and training techniques. In the next section, more details are provided. Fig. 3 presents the main difference between geometric methods and learning methods.

4. Deep learning in Visual Odometry

The primary goal of this paper is to investigate recent deep learning-based models found in the VO literature. Deep learning approaches have increased VO's accuracy and resilience, especially in difficult circumstances where standard methods may fail. The deep learning approach can be categorized into 4 subcategories: Supervised, unsupervised, self-supervised, and sim-supervised. In most literature, there is no distinction between self-supervised and unsupervised learning, and sometimes, it is confusing what differences there are. In unsupervised learning, no rule or guide is provided externally. Still, some constraints, limitations, or guidelines would be used in self-supervised learning to help the model train more appropriately. However, the literature ignores this slight difference, and these terms are used interchangeably. Moreover, as discussed later, the semi-supervised method is sometimes considered a hybrid method, which may confuse the readers. Therefore, in this paper, since there is no meaningful distinction between self-supervised and unsupervised learning by the authors, we combine these two subcategories into self-supervised as most of the recent papers use this term frequently.

4.1. Deep learning criteria

Efficient deep-learning models for camera pose estimation need to meet some requirements. Here are some of the criteria for deep learning methods for VO:

- Representation learning: Deep learning techniques for VO should be able to learn meaningful representations of image features that can be used for motion estimation. This may involve pre-training a network on a large dataset of images to understand feature representations relevant to motion estimation.
- End-to-end learning: Deep learning approaches for VO should be able to learn the mapping between visual sequences and motion from data directly without requiring intermediate stages like feature extraction or matching. By letting the network learn directly from the raw input, end-to-end learning can increase the accuracy and resilience of VO [5].
- Long-term motion estimation: Deep learning techniques for VO should be able to estimate motion over long-time intervals instead of short-term motion estimation that only considers the immediate past. Long-term motion estimation can improve the accuracy and robustness of VO in dynamic environments or where the agent moves slowly [5].
- Generalization: Deep learning techniques for VO should generalize well to new environments and conditions not seen during training. This requires the network to learn representations and mappings invariant to variations in lighting conditions, camera calibration, and scene structure.
- Computational efficiency: Deep learning techniques for VO should be able to operate in real-time or near real-time, particularly for applications. For example, low-latency motion estimation is crucial in robots or autonomous cars. This may involve designing lightweight and optimized networks for efficient computation.

4.2. Deep learning model architecture

This section briefly reviews different deep learning-based model architecture designs and training methods. Early methods mostly used the supervised training approach. However, recent studies prefer the self-supervised approach due to the difficulty of dataset annotation and the limitations, such as not covering many real-world scenarios. However, there are still some preferences to use a hybrid approach to take benefit of both training approaches [3]. Moreover, reviewing the literature, we observe that some papers introduce their work as "hybrid." This term has different meanings based on the authors. It is a general concept that includes other approaches than visual odometry. The third subsection quickly reviews the different approaches under the "Hybrid" term.

4.2.1. Supervised methods

Supervised methods are learning-based methods that rely on labeled training data, where each instance in the training set includes the input data and the corresponding desired output (ground truth). For the case of visual odometry, the input data would typically be images or image sequences captured by a camera (or cameras), and the corresponding output would be the relative motion (translation and rotation) between the images or image frames. Ground truth motion is usually obtained through high-accuracy motion capture systems, GPS, or other sensor modalities like LIDAR. The supervised learning algorithm learns a mapping from the input data to the desired output. Once trained, the model can predict the motion between new, unseen image pairs or sequences. The most common supervised learning models for visual odometry are Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), though other architectures like Transformers may also be used. There are different ways to frame the visual odometry problem in a supervised setting, such as:

- Frame-to-frame (or pairwise) VO: Here, the model takes a pair of images and outputs the relative motion between them. This straightforward setup may lead to error accumulation (drift) over long sequences.
- Sequence-to-sequence VO: The model takes a sequence of images and outputs the relative motion for the entire sequence. This is typically done using RNNs or other sequence processing models.

A key challenge in supervised methods for visual odometry is the need for large amounts of labeled training data, which can be difficult and costly to collect. It's also crucial that the model generalizes well to new, unseen environments, which is a significant challenge in machine learning. Another important aspect is that supervised learning methods only predict what they were trained for. Unlike methods like SLAM, if the model were trained only for estimating relative motion, it would not directly provide a global position or build a map of the environment. However, the VO estimates can be integrated with other sensor data or methods to achieve these tasks.

4.2.2. Self-supervised (Unsupervised) methods

Unsupervised VO is a technique that allows a computer vision system to estimate the motion of a camera in an unknown environment without using external sensors or prior knowledge of the scene. The model is trained on data without explicit supervision or labels in unsupervised VO. Instead, the model learns from the data's inherent structure by extracting patterns and regularities through unsupervised learning methods. The most common approach for unsupervised VO is to use deep neural networks to learn the feature representation and motion estimation jointly. The input picture is encoded to a feature representation, and the feature representation is decoded to the predicted camera motion. During training, the model takes a sequence of images as input and predicts the camera's motion between each pair of frames. The predicted motion is then compared with the ground truth motion obtained from a different source (such as a GPS or a motion capture system). The model's parameters are then tweaked using an optimization process, such as stochastic gradient descent, to minimize the discrepancy between anticipated and ground truth movements. This process is repeated for many iterations until the model's predictions are accurate enough for the given task. Once trained, the model can estimate the camera's motion in a new, unseen environment, using only the input image sequence.

4.2.3. Hybrid methods

Hybrid visual odometry (HVO) methods generally refer to approaches that combine multiple techniques or algorithms to enhance the precision and robustness of VO systems. There is no specific definition for the hybrid. Therefore, much research can be classified as a hybrid method. Some common examples of hybrid methods in VO include:

- Fusion of sensor modalities: This involves combining data from different sensors, such as cameras, IMUs, and LiDARs, to obtain a complete and more accurate estimate of the camera's motion.
- Feature-based and direct methods: Feature-based algorithms extract and match visual characteristics across successive frames to estimate camera motion, whereas direct methods employ complete picture information. Hybrid methods can use both approaches to benefit from their strengths and overcome weaknesses.
- Global and local optimization: Global optimization techniques, such as bundle adjustment, aim to jointly optimize camera motion and 3D structure over the entire sequence of images, while local optimization methods, such as visual-inertial odometry, optimize camera motion over short time intervals. Hybrid methods can use both approaches to improve the overall accuracy and efficiency of the VO system [3].
- Online and offline methods: Online methods estimate camera motion in real-time as new frames become available. In contrast, offline methods process the entire sequence of images offline to obtain a more accurate estimate of camera motion. Hybrid methods can combine online and offline processing to achieve real-time performance and high accuracy.

Despite deep learning techniques that have shown promise in improving the accuracy and robustness of VO, particularly in challenging environments, many research gaps should be addressed. The effectiveness of these techniques depends on several factors, including the training dataset's quality and size, the network architecture's design, and the optimization of training parameters.

5. Pose estimation

To estimate the pose, techniques are used individually or in combination to reach others. The most common methods to help better pose estimation are semantic segmentation, depth estimation, optical flow, scale estimation, and attention technique. In this section, we categorize the recent research based on their usage of information techniques to better understand the advancements in each method and find the research gap.

5.1. Optical flow

Optical flow is a visual odometry computer vision technology that estimates the mobility of an observer (such as a robot or self-driving automobile) by evaluating the movement of features in a succession of photographs. Optical flow detects patterns in pixel movement between two successive frames of a video stream. It assumes that a pixel's brightness stays constant over time and utilizes this assumption to determine its velocity by evaluating how its brightness varies between frames [6]. Optical flow may be employed in visual odometry to estimate a vehicle's motion by monitoring the movement of elements in the surroundings, such as edges, corners, or

other distinguishing points. It is feasible to estimate the observer's 3D mobility concerning the surroundings by examining the optical flow of these characteristics throughout a succession of photos. Optical flow is a robust approach that offers precise real-time motion estimates, even in difficult situations where other sensors like GPS or IMU may not perform well [6]. However, it is susceptible to noise, changes in lighting conditions, and other factors that can degrade the image stream's quality. In this section, the methods that used this technique are reviewed. However, like other methods, optical flow might also be used with other techniques to enhance the pose estimation accurately.

To train the ideal visual characteristics and predictor for ego-motion detection, old methods employed CNNs. Inspired by deep network developments and earlier work on VO learning approaches. They used dense optical flow and global information to estimate objects' scale and pose in noisy environments. They demonstrated on publicly accessible datasets that their technique is resilient to distortion, brightness, and contrast irregularities. Structure from motion (SfM) is described as a learning challenge. From sequential, unrestricted picture pairings, they train a neural network end to end to calculate depth and camera motion. The heart of the design, an iterative network that may enhance its predictions, comprises numerous stacked encoder-decoder networks. The first encoder-decoder block estimates the optical flow. Then, the following encoder-decoder estimates depth. Finally, a fully connected network estimates the pose. These 3 consecutive stages are repeated 3 times in a row. A training loss based on spatial relative differences, which produces more precise and reliable results, is an essential part of the methodology. DeMoN picks up the idea of matching and can generalize to structures not seen during training.

The learned depth produces a detailed 3D map. Curriculum Learning (CL) is a method of training a model that is inspired by human and animal cognition. Teed and Deng [7] present RAFT, a deep network design that improves optical flow precision. It records per-pixel attributes, creates multi-scale 4D correlation volumes for all pixel pairs, and updates a flow field regularly via a recurrent unit that searches up correlation volumes. On KITTI, RAFT lowers F1-all mistake. It also has excellent cross-dataset generalization, a short inference time, a fast training speed, and a large parameter count.

An end-to-end network for estimating camera ego-motion is presented by Huang et al. [8]. The network models sequential dynamics and learns the latent space of optical flow. In a self-encoding (self-supervised) manner, they compute the optical flow field of successive images and extract the latent optical flow representation. The 6-dimensional pose vector's regression is calculated using the extracted sequential optical flow latent space. The proposed model is tested on the KITTI and Malaga datasets. Al Hasib et al. [9] proposed the auxiliary task guiding (ATG) method. This method uses estimated gradient uncertainty for a mini-batch of data to guide an appropriate auxiliary task. It also uses optical flow subspace learning as another auxiliary task. The ATG approach provides a probabilistic fitness factor of the auxiliary task gradient for each shared weight at each mini-batch gradient descent training step to steer the main task. They demonstrated that this component involves task-specific confidence in manipulating ATG effectively. Monocular visual odometry is used to test the procedure. Monocular VO has been solved with a basic CNN architecture on the KITTI VO dataset.

Using scene-independent geometric calculations and BaXian inference, Li et al. [10] present a real-time adaptation structure. They use optical flow and depth estimation for estimating pose. Using a continuously learning optical variance, a differentiable Gauss-Newton network is also employed for additional depth and posture improvement. Li et al. [11] provide a technique that uses two camera position prediction models to handle the data from paired shots and a brief succession of images. A transformer-based supplementary pose generator, which builds a geometry model across a small period for a picture series, adopts a transformer-like structure (TAPE). A flow-to-flow pose estimator (F2FPE) is presented to exploit the connection among paired pictures. The two models are limited through a straightforward but efficient coherence loss in training. Scene structure is used for 3D motion prediction in pose estimation methods, which reduces resilience and makes cross-dataset generalization harder. On the other hand, structure from motion methods use optical flow to predict 3D motion and estimate depth, which is highly dependent on optical flow accuracy. Direct techniques have been developed using normal flow picture gradients to decouple 3D motion from depth estimates. According to the cheirality (depth positivity) requirement, Parameshwara et al. [12] network's NFlowNet for normal flow estimation is employed to calculate the 6-DoF. A differentiable cheirality layer formulated the optimization problem, allowing for end-to-end learning. On the datasets KITTI, TartanAir, and TUM-RGBD, they contrast their method with current cutting-edge methodologies.

[13] Cao et al.'s study [13] proposed a brand-new optical flow network (PANet). PANet predicts optical flow into discrete position space using optical flow probability volume, converts it to optical flow, and then computes it. The extracted flow is then used with DepthNet output to estimate the pose. Various sampling techniques, ego-motion initiation, adjusting the dynamic filtering factor, and weighting improve the bundle adjustment module for self-supervised training. A new normalized photometric loss function improves depth estimation. A way to enhance learning-based VO is to combine it with geometric-based methods. In this way, by matching geometrically triangulated depths to scale-consistent deep depths, Zhan et al. [14] solve the scale-drift issue and present a method to sample high-quality correspondences from deep flows and recover correct positions. [16] Chen et al. [15] estimate pose using per-pixel optical flow correspondence. Iterative refinement increases optical flow matching confidence, camera posture, and depth. The suggested method outperforms SuperPoint and SuperGlue in terms of in-depth and camera position accuracy. [18] A sparse optical flow network-based VO system is presented in [16] research by Liu et al. They created a pipeline using a consistency verification module and a Perspective-n-Point (PnP) solver. The suggested approach generalizes to real scenarios: the developed network's improved generalization capacity and the capacity to learn minimal datasets [20].

5.2. Depth estimation

Depth estimation is determining the relative depth or distance of objects within a scene from a sequence of images captured by a camera or multiple cameras. Visual odometry is essential in various applications such as robotics, autonomous vehicles, augmented

reality, and 3D scene reconstruction. [21] Li et al. [17] suggest an online meta-learning technique that allows VO networks to adapt to new environments continually. Rich geographical information has been aggregated using ConvLSTM in the past, and feature distributions have been aligned online using several methods. They used DepthNet to extract depth information, helping PoseNet, and finally used MaskNet to pixel mask based on wrapping residuals.

Liu et al. [18] introduced a deep learning-based RGB-D visual odometry system that makes use of an unsupervised end-to-end training technique and a dual-stream structure (RGB Stack and Depth Stack) using RGB pictures and depth map as inputs. The extracted features through dual streams are then fed into the LSTM network to estimate 6-DoF. To increase the system's positioning accuracy, Zhang et al. [19] built an unsupervised pose correction network using the resilience of deep learning and the multiview geometry limitations of the traditional stereo VO system. To improve the accuracy of conventional stereo VO placement, the pose correction network regresses a pose correction that causes positioning inaccuracy because it violates modeling assumptions. The ground truth poses dataset is not used to train the pose correction network. The posture correction network also produces a depth map and an explainability mask. DOC [20] is a novel deep online correction (DOC) framework for monocular visual odometry, combining convolutional neural networks and deep learning frameworks. Self-supervised CNNs are used first to create depth maps and starting postures. Second, CNN pose predictions are made even better by reducing photometric errors through gradient updates of poses during inference stages.

[26,27], Scale-consistent prediction is possible at the inference phase according to Bian et al.'s monocular depth estimation method SC-Depth [48]; their contributions include hybrid pseudo-RGBD SLAM, geometry consistency loss, and self-discovered mask. They provide an exhaustive ablation analysis to demonstrate the efficiency of each component, and they display excellent depth estimate outcomes in the KITTI and NYUv2 datasets. They easily integrate their monocularly trained deep networks into the ORB-SLAM2 for more reliable and precise tracking. They offer a variety of demos for qualitative assessment. The study by Zhao et al. [21] addresses a crucial issue of scale discrepancy for joint depth-pose acquisition. A unique approach is proposed that separates scale from network estimate. It can reconstruct the relative posture by directly calculating the basic matrix from the dense optical flow output and utilizing a two-view trigonometry unit to get a precise three-dimensional structure.

5.3. Semantic segmentation / integration

Semantic integration in visual odometry enhances VO algorithms with an understanding of the scene's context by recognizing the meaning and role of various objects within it. Traditional VO methods focus primarily on tracking geometric features or optical flow in image sequences to deduce the camera's trajectory. However, adding semantic information allows VO systems to perceive the environment more comprehensively, significantly improving the accuracy and robustness of pose estimation. Semantic segmentation has garnered interest among researchers as a method to enrich the scene context for pose estimation networks. It helps the system differentiate between the ground, static objects, and dynamic entities. Combining semantic segmentation with optical flow has shown promise in enhancing pose estimation, indicating a substantial research opportunity for developing more precise models. Dynamic environments, characterized by moving objects or people, pose significant challenges for VO. The motion of these elements can disrupt both the camera's motion estimation and the reconstruction of the scene's 3D structure. Moreover, moving objects can induce appearance changes in the scene, such as variations in lighting or texture, complicating the estimation process further. Below are the benefits of Semantic Integration in VO applications:

- Robust Feature Tracking: Semantic integration enables VO systems to selectively track features associated with stable, static environmental elements like buildings or roads. It helps ignore the features of dynamic and potentially disruptive elements, such as moving vehicles or pedestrians.
- Improved Data Association: Semantic integration facilitates more reliable feature matching by associating observed features with specific known object types, which is especially beneficial in complex environments.
- Handling Dynamic Environments: Semantic VO can effectively differentiate between static and dynamic scene components, vital for accurate navigation in environments where objects are in motion.
- Scale and Depth Estimation: Semantic information provides crucial context for estimating the scene's scale and depth. For instance, recognizing the standard size of a car or a pedestrian can inform the system's scale inference, particularly in monocular setups.
- Outlier Rejection: Semantics enhance the ability to detect and reject outliers by ignoring features that do not match the anticipated behavior of their semantic category.
- Long-term Operation: Semantic integration proves advantageous in consistently changing environments, such as those affected by construction or natural growth, by enabling the VO system to adapt to these changes over time.

Semantic segmentation of ground images is used to autonomously global localize planetary robotic explorers in harsh and GPS-denied situations. It registers saliency maps from a trained convolutional neural network to projected terrain elevation maps in the rover's general area of operation. It describes DeepLab V3+ and analyses the suggested method's performance using NASA's Perseverance rover's Martian landscape dataset [22]. A VO method [23] proposed in this work uses an image's dynamic elements by fusing semantic segmentation with the optical flow—the filter-moving object. Therefore, using semantic segmentation, moving objects are first found. An item is classified as dynamic and combined with other dynamic objects to generate a dynamic mask if it has several pixels with irregular optical flow. The camera's ego-motion is then calculated by only utilizing the static portions of the remaining picture and suppressing its dynamic portions. They use a geometric technique based on optimization as opposed to the other well-known deep learning-based VO methods to attain excellent performance. In other words, maximizing the correspondences

discovered using the dynamic mask and optical flow consistency creates the camera's ego motion. By utilizing the approximative of the actual ground plane's planarity, Muller and Smith [24] proposed a pose estimation model that uses the semantic segmentation method with an optical flow network to identify the ground over consecutive images and then RANSAC to choose the best fit. The pseudo labels they acquire by decomposing to a ground-relative posture can be utilized for direct monitoring. Compared to more sophisticated techniques that require learning relative posture and depth estimates, this straightforward geometric model is competitive for visual odometry.

5.4. Scale estimation

Scale estimation in VO is estimating the scale factor of the camera's motion in the scene. The scale factor is crucial because it determines the real-world units of the estimated 3D structure of the scene, such as the distance between objects or the size of objects. The scale estimation problem arises because VO algorithms estimate the camera's motion based on the image sequence's visual features without prior knowledge of the real-world scale of the scene. This means that the VO algorithm can only estimate the relative scale of the scene rather than the absolute scale. It is one of the significant challenges in pose estimation. The distance of the objects plays a considerable role, usually negatively, in determining the camera movement. Mainly, if an external database is used to help the model identify the type of objects, some research tried to focus on scale estimation to reduce errors in the ego-motion estimation to help the SLAM 3D model generate the camera environment.

In contrast to competing self-supervised approaches, the modifications some methods provide depth maps that are both numerically and qualitatively superior. These advancements include a minimal reprojection loss, a full-resolution multi-scale sampling technique, and an auto-masking loss to overlook training pixels that do not adhere to the camera motion assumptions. SAEVO [25] is a Siamese autoencoder visual odometry system that blends Siamese networks with an auto-encoder for multi-scale matching to estimate ego-motion. Compared to cutting-edge techniques, it performs well regarding posture and depth accuracy. According to light depth architecture [26] a monocular VO approach can gauge geometric postures and depth concurrently from a series of photos without ground truth. The depth model is improved by adding Atrous Spatial Pyramid Pooling (ASPP), which can calculate geometric postures and scene depth. The output reveals that the lighter depth design delivers predictions more accurately than the encoder-decoder, and the pose model outperforms several unsupervised models. VRVO [27] is a framework for obtaining absolute scale from virtual data from contemporary simulation environments. Utilizing monocular actual images and stereo virtual data to build a scale-aware disparity channel, it then uses GAN to map images from both datasets into a single feature space. They propose a mutual reinforcement pipeline to solve suboptimality produced by the separate optimization backend and learning process. Their framework performs well on KITTI and vKITTI2. Chaoqiang Zhao et al. [28] apply geometric restrictions to improve the pose network's scale consistency, including a new scale-to-trajectory constraint and a loss function improvement. Deep direct sparse odometry (DDSO) is a new DVO architecture that embeds TrajNet to alleviate the disadvantages of the prior DDSO framework.

5.5. Attention-based VO

Attention-based visual odometry is a technique used in visual odometry that incorporates attention mechanisms to improve the accuracy and robustness of motion estimation. This method analyzes the image's most informative and salient parts while ignoring irrelevant or noisy regions. Attention mechanisms are typically implemented using neural networks that learn to identify the most informative parts of the image based on the task at hand. For example, a neural network can be trained to focus on regions with high texture or edges, which are likely to provide more accurate motion estimates. It can also incorporate recurrent neural networks (RNNs) to model temporal dependencies in the image sequence, allowing the system to track motion over extended periods. Combining attention mechanisms with RNNs can achieve high accuracy and robustness even in challenging environments with changing lighting conditions, occlusions, or other noise sources. Attention-based methods have been shown to outperform traditional visual odometry techniques in various scenarios, including indoor and outdoor environments and applications such as robotics, autonomous driving, and augmented reality.

[36] Zhu et al. [29] examine monocular VO using Deep Learning. DeepAVO's premise is that characteristics distinguish motion patterns. They use CNNs to create a four-branch network to learn 6DoF from optical flow. Channel-spatial attention improves feature selection—the proposed DeepAVO. The proposed method is tested on the KITTI dataset. Song et al. [30], ContextAVO's method employs local contexts to improve estimates from numerous optical flow snippets. The Context-Attention Refining component adaptively uses camera motion continuity and local contexts to improve current inference. They also use a multiple-length window to generalize the model and reduce its dependency on a fixed window. Their methodology is tested on outdoor KITTI, Malaga, ApolloScape, and interior TUM RGB-D datasets. Liu et al. [31] developed a confidence-based end-to-end VO model. It is divided into two stages: the baseline relative posture transformation being predicted using a confidence mask and determining the confidence level of the final 6-DoF estimate using spatial integrity of the path. The suggested approach's evaluation of the model on the KITTI demonstrates its usefulness in enhancing pose estimation accuracy and resilience. A Siamese autoencoder visual odometry system called SAEVO has been developed by Liang et al. [55]. To estimate ego-motion, Siamese networks and auto-encoder are combined. To align the extracted features, two unsupervised losses are used. The KITTI and CityScapes datasets have been used in experiments.

5.6. Direct methods

Han et al. [32] offer SuperPointVO, a lightweight stereo visual odometry system, using convolutional neural network feature

extraction. They eliminate redundant descriptor information and enhance expression capabilities by NMS and grid sampling. They test the system on the KITTI dataset, which is comparable to existing stereo SLAM systems. Kaygusuz et al. [33] propose MDN-VO, a deep learning-based VO, and confidence model, to estimate 6-DoF poses efficiently. It uses CNN and RNN to learn features and then uses a Mixture Density Network (MDN) to extract 6DoF. They tested the model on the KITTI and nuScenes datasets and presented pose and uncertainty estimation results. Their experiments show that the proposed model detects failures using predicted pose uncertainty. By choosing memory and fine-tuning postures, Xue et al. [34] suggested a novel deep visual odometry approach that considers global information. As a neuronal equivalent of memory, their architecture's Tracking, Remembering, and Refining modules gradually and adaptively preserve information from local to global. The method is tested on KITTI and TUM-RGBD datasets. It performs superbly under challenging situations like sudden movements and regions without texture.

Learning-based VO focuses on long- and short-term sequential learning schemes without considering the underlying rotation-translation ambiguity that causes VO drift. Lin et al. [35] created an unsupervised monocular VO framework to estimate 3D camera positions by separating rotation, translation, and scale estimates. They start with a cascade and a residual-based decoupled pose refinement approach for exact posture assessment. They then incorporate them into an unsupervised monocular VO framework that computes 3D camera locations by disentangling rotation, translation, and scale predictions. Using two neural network architectures based on the Early Fusion and Slow Fusion structure, Gilles and Ibrahimpasic [36] provide a different method based on deep learning for calculating ego motion using downward-facing cameras. Both networks are trained with a modified structural similarity index loss function (SSIM) and share a spatial transformer layer for image warping. [45]

6. Combination of techniques

Deep learning-based VO architectures leverage neural networks to extract features, estimate motion, and reconstruct views, facilitating precise pose estimation even in complex environments. These architectures are designed to handle various aspects of the VO pipeline, from direct pose estimation to integrating semantic information and attention mechanisms. In the following sections, we delve into six distinct architectures that exemplify the innovative approaches in deep learning-based VO, explaining the unique components and data flow processes of each. These architectures demonstrate the versatility and adaptability of deep learning techniques in enhancing the VO system's ability to understand and interpret visual data. This section shows various views of combining the aforementioned techniques and explains them in detail.

6.1. Pipeline 1

The architecture depicted in Fig. 4 represents a typical deep learning-based VO pipeline designed to estimate the camera's motion through a sequence of images. Here's a breakdown of the flow and components:

- Image Sequence: The process starts with images captured over time, which serve as the input data.
- Feature Extraction: This step involves analyzing the images to detect and describe salient points or features that can be used for tracking motion across frames. These features are typically corners, edges, or distinct textures easily identified across successive images.
- Outlier Rejection: Once features have been extracted, some may be erroneous or not correspond to actual points in the scene (outliers). These are identified and removed to improve the robustness and accuracy of the motion estimation process.
- Feature Matching/Tracking: The remaining features are matched or tracked across the image sequence. This involves identifying the same feature in subsequent images to understand how each point moves, crucial for deducing camera motion.
- Motion Estimation: With the matched features, the algorithm estimates the relative motion of the camera between frames. This typically involves solving for rotation and translation that best aligns the matched features between image pairs.
- Camera Calibration: Camera calibration information is used in parallel to feature matching. This information is crucial as it defines the camera's internal characteristics (like focal length and optical center) and is used to correct the images and measurements for any distortions.
- Scale Estimation: For systems where the scene's absolute scale is unknown (such as monocular VO), scale estimation is performed to convert the relative motion into real-world units.
- Bundle Adjustment: This step refines the camera's trajectory and the 3D structure of the scene by minimizing the re-projection error, which is the difference between the observed feature positions and the projected positions of the estimated 3D points.
- Pose Estimation: Finally, the camera pose (position and orientation) for each image in the sequence is estimated based on the refined motion and structure from the bundle adjustment.

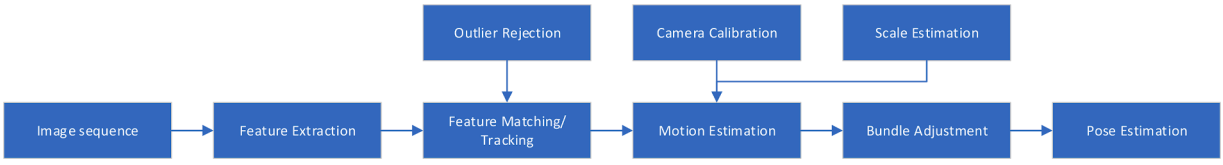


Fig. 4. Architecture of Pipeline 1 Pipeline 2.

Data flows sequentially through this pipeline, starting from raw images and culminating in an estimate of the camera's pose. At each step, the information is processed and refined, with some components like camera calibration providing the necessary parameters for interpreting the image data correctly. The entire process is underpinned by deep learning algorithms, which can optimize feature extraction, matching, and motion estimation tasks.

The architecture in Fig. 5 is a streamlined deep learning-based visual odometry system with three main components:

- Image Sequence: This is the input to the system, typically a chronological series of images captured by a camera. The image sequence captures the dynamic visual information as the camera moves through the environment.
- Deep Network: This central component is a deep neural network, which processes the image sequence. Depending on the network's architecture, the network is likely designed to perform a combination of tasks such as feature extraction, feature matching, motion estimation, and possibly even scale estimation. In deep learning-based VO, convolutional neural networks are often employed for their ability to extract spatial hierarchies of features hierarchically. The deep network is trained on a dataset of image sequences with known camera motions, enabling it to infer the camera's pose from visual information alone.
- Pose Estimation: The output of the system is the pose estimation. This stage translates the deep network's inference into a pose—comprising the position and orientation of the camera at each point in the sequence. The pose estimation is crucial for applications such as navigation, mapping, and augmented reality, where understanding the camera's movement through space is essential.

The data flow in this architecture is sequential and direct, moving from the image sequence through the deep network and finally to the pose estimation. Unlike more complex VO systems, there are no additional steps for outlier rejection or explicit scale estimation, which suggests that the deep network handles these aspects internally. The simplicity of this architecture implies a highly integrated system where the deep network plays a significant role in processing and interpreting visual data for VO.

6.2. Pipeline 3

The architecture shown in Fig. 6 outlines a multi-faceted approach to deep learning-based visual odometry (VO). Here's an explanation of the components and data flow:

- Image Sequence: The process starts with a sequence of images used as the raw data input. These images capture the scene as the camera moves through an environment.
- Depth Map/Flow: A depth map or depth flow is generated from the image sequence. A depth map provides per-pixel distance information from the camera's perspective, while depth flow refers to the pattern of depth change across the image sequence.
- Optical Flow: In parallel, optical flow is calculated from the image sequence. The apparent motion of objects, surfaces, and edges in a visual image is generated by the relative motion of an observer (the camera), and the scene is quantified by optical flow.
- Pose Network: Simultaneously, a pose network deduces the camera's motion (pose) from the image sequence. This network is typically a specialized neural network trained to estimate the camera's position and orientation.
- Deep Network (CNN/RNN/FNN): The depth map/flow, optical flow, and the output of the pose network are then fed into a deep neural network, which could be a CNN, RNN, or FNN. This network deeply processes these multiple inputs to understand the scene's structure and motion dynamics.
- Pose Estimation: Finally, the deep network produces a pose estimation. This represents the estimated trajectory of the camera, detailing its position and orientation over time, which is essential for tasks such as autonomous navigation, 3D reconstruction, and augmented reality applications.

The data flow in this architecture is more complex than a simple sequential flow, as it involves parallel processing paths that converge in the deep network. Each processing path extracts different information from the image sequence, and the deep network integrates these to estimate the pose accurately. This integrated approach allows the system to leverage multiple aspects of the visual data, potentially leading to more robust and accurate VO.

The architecture in Fig. 7 represents a deep learning-based visual odometry pipeline with several key components:

- Image Sequence: This is the primary input to the system, comprising a series of images captured over time that will be used to infer the camera's motion.
- Depth Net: The first stage in the process is the Depth Net, a neural network designed to estimate the depth of each pixel in the images. This depth information is crucial for understanding the 3D structure of the environment from 2D images.



Fig. 5. Architecture of Pipeline 2.

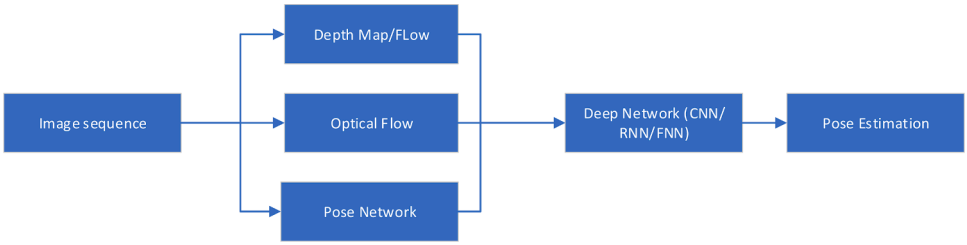


Fig. 6. Architecture of Pipeline 3 Pipeline 4.



Fig. 7. Architecture of Pipeline 4 Pipeline 5.

- Optical Flow: Subsequent to depth estimation, optical flow is computed. Optical flow algorithms estimate the motion of objects between two frames of the image sequence based on the varying intensity patterns. This motion information is critical for tracking how objects move relative to the camera.
- Deep Network (CNN/RNN/FNN): The depth information and optical flow are fed into a more complex deep neural network, a CNN, RNN, or FNN. This network integrates the information provided by the previous components to understand the scene's geometry and the camera's motion.
- Pose Estimation: The final output of the system is the pose estimation. This component takes the consolidated data from the deep network to calculate the camera's position and orientation (pose) through space over time.

Data flow is sequential but hierarchical, with each stage building upon the previous one's outputs. The process starts with primary image data, progressively adds layers of spatial and motion information through the depth and optical flow calculations and culminates in a complex interpretation of camera motion with the deep network, leading to the final pose estimation. This pipeline allows the system to create a detailed and dynamic understanding of the camera's trajectory and orientation in space.

The architecture depicted in Fig. 8 represents a deep learning-based visual odometry (VO) system designed to estimate camera pose using semantic information and attention mechanisms. The flow of data and the function of each component are as follows:

- Image Sequence: The system starts with an input sequence of images, which captures the environment through which the camera moves.
- Semantic Segmentation: The first processing step involves semantic segmentation, which classifies each pixel in the images into predefined categories based on the features present. This step is critical for understanding the scene's context, such as identifying roads, buildings, and other objects.
- Optical Flow: The system computes optical flow from the image sequence parallel to semantic segmentation. Optical flow algorithms determine objects' apparent motion patterns between consecutive frames, which is used to infer the camera's movement relative to its surroundings.
- Attention: Another parallel process involves an attention mechanism. This part of the system focuses on the most relevant features or regions in the images for the task at hand, which can enhance the accuracy and efficiency of the pose estimation by prioritizing important information over less relevant data.
- Deep Network (CNN/RNN/FNN): The outputs from semantic segmentation, optical flow, and attention are then integrated by a deep neural network, which could be any combination of a CNN, RNN, or FNN. This network extracts complex features and temporal dependencies from the data.
- Pose Estimation: Finally, the pose estimation is derived from the processed information by the deep network. This step calculates the camera's position and orientation within the environment based on the semantic and motion cues processed by the system.

The data flows through the system in a somewhat parallel and convergent manner, with semantic segmentation and optical flow

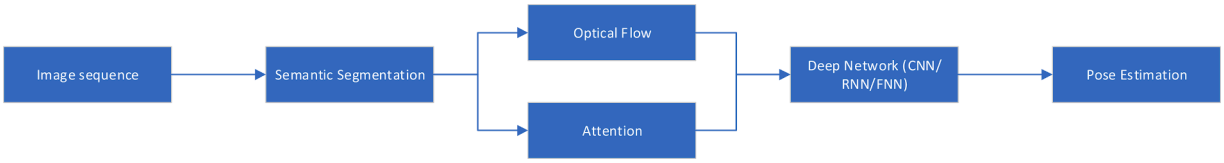


Fig. 8. Architecture of Pipeline 5.

providing spatial and motion cues. At the same time, attention focuses processing power on the most critical elements. The deep network then synthesizes these to determine the camera's trajectory and orientation, providing the final pose estimation output. This architecture is indicative of an advanced VO system that utilizes the rich semantic context of the scene, which can lead to more robust and accurate navigation and mapping.

6.3. F. Pipeline 6

The architecture depicted in Fig. 9 represents a complex deep learning-based visual odometry pipeline. Here's how it works:

- Image Sequence: The system takes an image sequence as input. This sequence is a series of images captured over time that shows the scene from the camera's perspective as it moves through the environment.
- Deep Network: This neural network processes the image sequence to extract features and learn the underlying patterns. It serves as the system's backbone, performing critical tasks such as feature extraction, initial motion estimation, and possibly depth estimation.
- Generative Network: The output of the deep network is passed to a generative network. This could be like a Generative Adversarial Network (GAN), which can generate new data similar to the original input images. This can be used for tasks such as view synthesis, where new viewpoints are generated based on the learned representations, or for augmenting the dataset to improve the system's robustness.
- View Reconstruction: Following the generative network, the system performs view reconstruction. This step involves creating a coherent and continuous representation of the scene, such as a 3D model, from the synthesized views or the enhanced dataset provided by the generative network.
- Discriminative Network: Finally, a discriminative network is applied. This could be a network trained to differentiate between different types of motion patterns or to refine the pose estimation. The discriminative network makes the final decision on the camera's pose based on the reconstructed views, ensuring that the pose estimation is accurate and consistent with the physical world.

The data flows sequentially from raw images through feature extraction and augmentation to reconstruct views and, finally, to precisely estimate the camera's pose. This architecture shows an advanced VO system that uses deep learning not only for direct pose estimation but also for enhancing the input data and refining the output, potentially increasing the accuracy and reliability of the VO system.

7. Deep learning-based visual odometry model accuracy evaluations and benchmarking

The accuracy of deep learning-based Visual Odometry models is typically evaluated using a variety of benchmarks and metrics that assess how well a model estimates the camera's trajectory and orientation. Here's a discussion of standard practices for evaluating VO model accuracy, including benchmark datasets, evaluation metrics, and examples.

7.1. Benchmark datasets

VO models are often benchmarked against standard datasets with ground truth poses. Examples of these datasets include:

KITTI Vision Benchmark Suite: One of the most widely used datasets for evaluating VO, which provides a real-world setting with various driving scenarios.

TUM RGB-D Dataset: Contains RGB and depth images captured with a Microsoft Kinect sensor, providing ground truth trajectories for indoor scenes.

EuRoC MAV Dataset: Offers data from Micro Aerial Vehicle (MAV) flights, including stereo images and IMU data in indoor and outdoor environments.

7.2. Evaluation metrics

The performance of VO models is quantified using specific metrics that can compare the estimated trajectory to the ground truth. The most common metrics are:

Absolute Trajectory Error (ATE) measures the difference between the estimated trajectory and the ground truth at each time-stamp. It is computed as:



Fig. 9. Architecture of Pipeline 6.

$$ATE = \sqrt{\frac{1}{N}\sum_{i=1}^{N} ||P_i - \widehat{P}_i||^2} \tag{1}$$

Where P_i is the ground truth position, $\widehat{P}_i$ is the estimated position, and N is the number of positions.

Relative Pose Error (RPE): It measures the local accuracy over a fixed interval, assessing the drift in the pose estimation. It can be computed for translation and rotation separately:

$$RPE_{trans} = \sqrt{\frac{1}{N-d}\sum_{i=1}^{N-d} ||t_{i+d} - Q_i t_i - \widehat{t}_{i+d} - \widehat{Q}_i \widehat{t}_i||^2} \tag{2}$$

$$RPE_{rot} = \sqrt{\frac{1}{N-d}\sum_{i=1}^{N-d} \left|\left|\log\left(Q_i^{-1} Q_{i+d} \widehat{Q}_i^{-1} \widehat{Q}_{i+d}\right)^V\right|\right|^2} \tag{3}$$

Where $\boldsymbol{t}$ and $\boldsymbol{Q}$ are the translation and rotation components of the ground truth pose, $\widehat{\boldsymbol{t}}$ and $\widehat{\boldsymbol{Q}}$ are the estimated components, $\boldsymbol{d}$ is the interval length, and $\log(.)^V$converts a rotation matrix to its axis-angle representation.

Scale-Drift Evaluation: This is particularly important for monocular VO systems, which cannot directly observe the scale of the scene. It measures how much the estimated scale drifts from the true scale over time.

Example Evaluations: Consider a VO model tested on the KITTI dataset. The model's estimated trajectory is compared against the ground truth using ATE and RPE. If the ATE is found to be 5 m over a 500-m trajectory, this indicates a 1 % drift from the ground truth. Similarly, if the RPE for rotation over a 10-s interval is 0.01 radians, it suggests a small rotational drift, indicating high local accuracy.

Benchmarking Practices: When benchmarking VO models, it is important to ensure that:

- The model is tested on a variety of sequences to ensure generalizability.
- Evaluation metrics are applied consistently across different models for fair comparison.
- External factors such as camera calibration and synchronization issues are considered, as they can affect the accuracy.

Benchmarking allows researchers and developers to objectively compare the performance of different VO models, thereby facilitating the development of more accurate and robust systems. By continuously testing against these benchmarks, the field can track the progress and improvements in VO techniques.

8. Datasets

In this section, the most common datasets that researchers use in deep learning-based VO are presented.

8.1. TartanAir [37]

TartanAir is a large-scale, photorealistic synthetic dataset for training and evaluating computer vision algorithms, particularly those related to autonomous navigation and robotics. It was created by researchers at Carnegie Mellon University's Robotics Institute. The dataset includes indoor and outdoor environments with diverse lighting conditions, weather patterns, and terrain types. The dataset features high-resolution RGB images, depth maps, segmentation maps, dense ground-truth camera poses, and trajectory data. One of the unique features of TartanAir is its ability to generate synthetic data with a high degree of realism, using advanced physics and rendering engines. This allows researchers to train and test computer vision algorithms in various challenging scenarios without needing expensive or time-consuming real-world data collection. TartanAir has been used in several research projects and competitions, including the RoboCup 2D and 3D simulation leagues, and has become a popular benchmark for evaluating the performance of autonomous navigation and robotics algorithms.

8.2. CrowdDriven [38]

The location and orientation of a picture (or a sequence of images) inside a known scene are crucial components of many robotics and computer vision applications, from augmented and virtual reality systems to self-driving automobiles. Techniques for visual localization should function consistently and stably under various circumstances, including seasonal, weather, lighting, and artificial variations. The community has made significant progress on these datasets since they were created. Recent benchmarking attempts represent this by providing photographs under various situations. However, they are confined to a few geographical areas and frequently documented using just one instrument. With an emphasis on the failure situations of existing algorithms, they suggest a new benchmark for visual localization in outdoor scenes utilizing crowd-sourced data to span various geographical locations and camera devices. Studies using cutting-edge localization techniques reveal that their dataset is challenging, with all tested algorithms failing on their most difficult portions. They share the dataset with the tools used to create it, allowing quick and easy 2D correspondence annotation to produce reference poses.

Table 1
A comprehensive summary of deep learning-based VO methods.

Paper	Input Data	Architecture Network	Techniques						Estimation Metric		Learning Model		Dataset
			Direct	Semantic Segmentation	Scale Estimation	Attention	Optical Flow	Depth Estimation	Location	Pose	Supervised	Self-supervised	
[22]	Monocular Camera	CNN		X					X		X		Martian landscape - NASA
[32]	Stereo Camera	keypoints and descriptors extractor, BA optimization	X							X		X	KITTI
[43]	Monocular Camera	Attention map based on OF (FlowNet2) and semantics (DeepLab-v3)				X	X			X	X		KITTI
[7]	Monocular Camera	Feature & Context Encoders					X				X		KITTI
[8]	Monocular Camera	OF, RNN, CNN Encoder					X			X	X	X	KITTI
[33]	Monocular Camera	CNN, RNN, MDN	X							X	X	X	KITTI, nuScenes
[34]	Monocular Camera	Tracking, Remembering, Refining	X							X	X		KITTI, TUM-RGBD
[29]	Monocular Camera	Four-branch network, channel-spatial attention, OF				X	X			X	X		KITTI
[30]	Monocular Camera	Multi-length window,				X	X			X	X		KITTI, Malaga, ApolloScape, TUM RGB-D
[9]	Monocular Camera	Probabilistic auxiliary task guidance					X			X	X		KITTI
[17]	Monocular Camera	ConvLSTM						X		X		X	KITTI, TUM-RGBD
[18]	Monocular Camera	Dual stream RGB and depth betwork						X		X		X	KITTI
[35]	Monocular Camera	Separate netowrk for rotation and translation	X							X		X	KITTI
[19]	Stereo Camera	Correction Network						X		X		X	KITTI
[25]	Stereo Camera	Siamese network, Auto-encoder			X			X		X		X	KITTI, CityScapes
[20]	Monocular Camera	D3VO						X		X		X	KITTI
[36]	Monocular Camera	Early Fusion and Slow Fusion	X							X		X	Own dataset
[10]	Monocular Camera	FlowNet2, Depth Net, CNN					X	X		X		X	KITTI, TUM
[11]	Monocular Camera	TAPE, F2FPE					X	X		X		X	KITTI
[31]	Monocular Camera	CNN, GRU				X				X		X	KITTI
[28]	Monocular Camera	semantic segmentation			X			X		X		X	KITTI
[12]	Monocular Camera	PoseNet, NFlowNet					X			X		X	KITTI, TartanAir, and TUM-RGBD
[24]	Monocular Camera	Non-Differentiable Homography Estimation		X			X			X		X	KITTI
[44]	Monocular Camera	Two stages: 1. Local time window 2. Refines the poses with a “global” loss	X							X		X	KITTI and TUM RGB-D
[45]	Monocular Camera	Pseudo RGB-D, Depth Network, RGB-D SLAM						X		X		X	KITTI and TUM RGB-D

8.3. KITTI VO/SLAM benchmark [39]

One of the most well-known public datasets for testing VO and visual SLAM algorithms is the KITTI VO/SLAM benchmark, which was developed during outdoor automobile driving activities. It contains 22 picture sequences, 11 of which (Sequences 00–10) are connected to precise ground truth. Sequences 11 through 21 are the remaining ten sequences, and they are solely given the raw sensor data without any ground truth or GPS. It is quite difficult for monocular VO algorithms to process this dataset since it was captured at a very modest picture capture rate (10 frames per second) while driving through metropolitan regions with plenty of dynamic objects and at speeds of up to 90 km/h. Even though the dataset includes imagery from all of their cameras, only the photos from the left camera (the Point Grey Flea 2 camera) are utilized in their monocular VO tests. Applications for mobile robots, such as autonomous driving, are represented by the KITTI dataset.

8.4. Málaga urban dataset

The Málaga urban dataset [40] is gathered while driving through metropolitan areas. It offers data from LiDAR, GPS, and other sources coupled with 20 Hz stereo imaging. They only utilize the left camera's photos in their study and use them only to test trained models already taught. Its photographs are cropped and scaled to the KITTI image size since their image size differs from that of the KITTI.

8.5. Cityscapes dataset

The Cityscapes dataset [41] aids in interpreting urban scenes using extensive semantic segmentation and labeling. Additionally, it is obtained when navigating urban areas. It includes several stereo video clips recorded in various cities to estimate VO. Like the Málaga dataset, the Cityscapes dataset is exclusively used to test models.

8.6. The European robotics challenge (EuRoC) (Indoor)

A dataset including visual inertial data obtained with an MAV was released in 2016 [42]. It may be used quickly for SLAM, 3D reconstruction, and visual-inertial odometry. Raw readings of an Inertial Measurement Unit (IMU), stereo images, ground truth, and sensor calibration of 11 sequences are provided for two settings, a factory hall, and a Vicon room. It captures motion-blurred, underexposed photographs and high-quality ones of a flying robot performing various actions, from deliberate moves to agile flight, which may provide more information about the dataset and setups [80]. They mostly employ the five Machine Hall sequences since it is larger than the Vicon Room. The first 200 photos from each series are discarded since they were used for visual-inertial calibration, and some of their views were restricted to the ground. An excellent illustration of indoor applications for flying robots may be found in the EuRoC MAV dataset.

9. Comparative study and discussion

Deep learning has revolutionized the field of visual odometry (VO) by introducing a plethora of techniques to enhance accuracy and robustness. Among the myriad of techniques, semantic segmentation has gained popularity due to its ability to distinguish and categorize different objects and structures within an image, aiding in discerning static elements from dynamic ones. By recognizing these different segments in sequential frames, the VO models can better determine the movement and position of the camera. Another significant advancement in VO is scale estimation. This technique addresses the challenge of determining the true movement scale between frames. Deep learning-based VO models can deduce the relative distance and orientation between different frames by accurately estimating scale, ensuring more precise trajectory estimation. To further refine the results, attention mechanisms are integrated into VO models. This allows the models to focus on the most relevant parts of an image sequence, ensuring that critical features are prioritized and less informative ones are downplayed. Incorporating optical flow and depth estimation in deep learning-based VO further augments the precision. Optical flow focuses on visual objects' motion patterns between consecutive frames. At the same time, depth estimation aims to capture the third dimension, providing insights into how far objects are relative to the camera. Some proposed methods use a combination of these techniques, and some only focus on a single one. The results show that the models that use multiple techniques to compensate for the drawbacks of others provide more robust and generalized results. However, the computational complexity of this approach is considerably higher and raises the concern of being applied for real-time application. Therefore, one of the important research gaps is proposing models with robust output that can be used in real-time applications. Moreover, enhancing each technique's performance will also help the pose estimation. For instance, optical flow methods suffer from light conditions, and the model performance degrades significantly in low-light environments. Proposing models that perfume in different light conditions can also generalize the proposed pose estimation models. Another limitation of the mentioned methods is the neural network's capacity for learning. CNN is widely used in computer vision and VO. Its capacity to learn after a certain point degrades to the extent that it no longer learns by adding more training data. Additionally, CNN removes much information that can be used to train the rest of the network. Using transformers instead of CNN or within CNN can compensate for the mentioned limitations and enhance the performance of the VO methods. A comparison becomes essential in understanding the applications and efficiencies of these techniques. Table 1 presents a comprehensive summary of each method in relation to deep learning-based VO models.

As observed in Table 1, we can identify each technique's strengths, limitations, and unique features, facilitating an informed

decision on the most suitable approach for any given application. Despite extensive research to enhance the performance of each model, there are still many research gaps that most of the papers mentioned. As a summary, the generally identified research gaps would be:

- Robustness of optical flow models to varying lighting conditions: Optical flow models are sensitive to illuminary change and low lighting conditions.
- Long-term consistency and drift correction: Despite much research, cumulative error is a big challenge in pose estimation. However, while many hybrid methods, including SLAM, are proposed to minimize this error, there is still a gap for more research.
- Handling occlusions and dynamic objects: A dynamic environment is a very challenging problem for pose estimation. The moving object significantly distracts models' prediction, and few research addresses this problem.
- Real-time performance and computational efficiency: Due to extensive computational needs for pixel-level analysis, most of the proposed models are not fast enough to be used in an immediate peace environment where the scene is rapidly changing due to the speed of the sensor platform or dynamic objects.
- Semantic-aware visual odometry: Few research has been conducted considering the context of visual odometry that uses a general semantic knowledge of the environment in which the odometry is performed.
- Integration with other sensors: Using a single modality to predict pose is cost-efficient and has the advantage of easy implementation. However, in recent years, the prices of other sensors have continuously decreased, allowing researchers to propose models with multiple inputs like LiDAR, IMU, etc.

10. Business adoption

Using visual odometry (VO), self-localization has found its footing in various practical applications, underscoring its pivotal role in modern technology and everyday life.

- **Autonomous Vehicles:** One of the most notable applications of VO-based self-localization is in the realm of autonomous vehicles. Understanding vehicle position and orientation in real-time is essential, whether passenger cars, drones, or underwater autonomous vehicles. Analyzing consecutive camera frames, these vehicles can remarkably precisely navigate urban jungles and uncharted territories.
- **Augmented Reality (AR) & Virtual Reality (VR):** AR and VR platforms leverage VO to enhance user immersion. In AR, for example, accurate self-localization allows for the seamless integration of virtual objects into the real world. VR ensures that the user's movement in the physical world is accurately mirrored in the virtual environment, making experiences more realistic and interactive.
- **Robotics**: Robots benefit immensely from VO-based self-localization, especially in dynamic and unpredictable environments. From robots in warehouses that need to navigate complex routes while avoiding obstacles to robots used in disaster response in environments where GPS might be unreliable, the ability to understand their position relative to their surroundings is crucial.
- **Space Exploration**: In environments where GPS is unavailable, such as outer space or other celestial bodies, rovers and landers employ VO for navigation. For instance, Mars rovers use visual odometry to navigate the Martian terrain, ensuring they do not get lost while executing their exploration missions.
- **Smartphone Applications**: Modern smartphones, with their advanced camera systems, also harness the power of VO. Features like live translation (wherein text from the real world is translated in real-time on the phone's display) or star map applications (that identify stars and constellations when pointed at the night sky) are made possible with precise self-localization techniques.
- **Sports and Filmmaking**: In sports analytics, accurate player and ball tracking have become invaluable for performance analysis and game strategies. Similarly, in filmmaking, VO helps in camera tracking, especially in scenes that rely heavily on CGI, ensuring flawless integration between real footage and computer-generated imagery.
- **Drone Navigation:** Drones utilize semantic VO for surveying, search and rescue, and delivery tasks. By recognizing and categorizing different terrains and obstacles, drones can adjust their flight paths autonomously for optimized routing and obstacle avoidance.
- **Historical Site Reconstruction:** In archaeology and historical research, semantic VO can assist in the digital reconstruction of sites. By identifying and segmenting ruins and landscape features, researchers can create more accurate 3D models of historical sites.
- **Assistive Technology for the Visually Impaired:** Semantic VO can be integrated into assistive devices to provide spatial information to visually impaired individuals. By identifying and describing the surrounding environment, these systems can guide users through unfamiliar spaces with audio cues.
- **Mapping and Surveying:** Geospatial companies use semantic VO to improve the accuracy of maps and 3D models of cities. The technology can create detailed and up-to-date maps by understanding elements like buildings, trees, and roads.

The versatility and accuracy of visual odometry-based self-localization underscore its growing significance. As technology continues to advance, it is evident that the scope of its applications will only expand, making self-localization an indispensable tool in various industries and endeavors.

11. Conclusion and future directions

Visual odometry faces substantial challenges in handling occlusions and dynamic objects in dynamic environments, and research in this area is crucial for making visual odometry systems more robust and reliable in real-world settings. Research focusing on optimizing the computational efficiency of visual odometry models is essential for real-time performance in applications like autonomous vehicles and robotics. This could involve developing hardware acceleration, algorithmic optimizations, or novel neural network architectures tailored for efficient real-time operation. Integrating semantic awareness into visual odometry is an exciting avenue for research, as it can lead to more contextually aware systems that better understand and navigate complex scenes. Despite the progress made in visual odometry, significant research gaps still need to be addressed. By addressing these challenges, researchers can continue advancing the visual odometry field and unlock its full potential in applications such as autonomous vehicles and robotics. While advancements in deep learning-based visual odometry have been remarkable, addressing these systems' limitations and challenges is crucial. First, the dependency on large and diverse training datasets for deep learning models is a significant hurdle. The models often fail to generalize to environments and situations that differ from the training data, reducing accuracy in unseen or dynamic conditions. Additionally, the computational intensity of deep networks can pose deployment challenges, particularly in real-time applications or on hardware with limited processing capabilities. Deep learning-based VO systems also require substantial power consumption, which can be a critical drawback for battery-operated devices like drones or mobile robots. These systems may struggle to maintain consistent performance when environmental conditions such as drastic lighting shifts or weather changes rapidly change. Another limitation lies in handling reflective, transparent, or texture-less surfaces, which can disrupt feature extraction and matching processes. Furthermore, the inherent issue of scale ambiguity in monocular VO systems remains a challenge that deep learning has not fully overcome, often necessitating additional sensors or methods to estimate scale accurately. The deep learning models can be black boxes, making diagnosing failures or understanding decision-making processes difficult. This lack of interpretability can be problematic in applications requiring high trust and safety levels, such as autonomous vehicles or assistive robotics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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